Unit 3 - Impact of IT & Information Systems
Assignment 4

Introduction

Scenario / Vocational Context: *Now that iTech Consultancy Company has begun its marketing phase, the Company also needs to complete internal research for the solutions it will offer to its first big client, DHL Logistics. As such your Manager has asked you to research and review a number of products and services that are new to the market, which may be suitable for DHL Logistics.*

Task 1

Task description - *Find and research at least four contemporary technologies that may be used by DHL to solve the process in the Case Study Provided. Your chosen technologies should contain the following types:*

- *ICT Hardware i.e. Systems, components and wired networks.*
- *Software i.e. Web tools, apps or computer programs with good integration tools.*
- *Wireless Technologies and Devices i.e. Routers and portable devices*
- *Operating Systems and Software Programming Languages*

(U4:P5) - Describe recent IT developments

Describe the key features of the technology

ICT hardware such as global positioning systems (GPS), is a piece of hardware that tracks the current position of the user anywhere around the globe, it does this by using data from at least four GPS satellites.

Generally, a GPS doesn't need the user to do anything to find out where it is. Instead it depends on telephone or internet connections. This means that the data that a GPS gathers is automatic, which reduces user interaction to a miniscule amount.

This means that the GPS is a convenient piece of technology that can be installed and not interrupt the employee. High end GPS use different technologies to enhance the usefulness of the GPS positioning information, making it much more useful and convenient for the driver of the vehicle.

Tracking applications, or ATS (applicant tracking system), is an application that can be implemented or accessed online in the business. What I mean by this is that, the basic function of an ATS is to give a central location and database for a company's recruitment efforts.

This means that the data is collected from internal applications via the ATS front-end, which is usually located on the company website or it could be extracted from applicants on job boards. Which basically means that it could either be found online or it could be taken from people who applied on the job boards.

ATS can have other features, such as individual applicant tracking, requisition tracking, automated resume ranking, customized input forms, pre-screening questions and response tracking. This ability in a piece of technology is useful because it makes it unique and very useful for a business at the same time.

Portable devices such as tablets, smartphones and even smart watches are becoming increasingly common, giving these mobile devices more versatility. This means that this type of technology can be used to increase the efficiency of a business.

With these devices, aspects such as networking/communication, development, marketing and commerce have a possibility of being massively improved within a business just via mobile technology, due to the important data that can be gathered from the usage of mobile technology.

Operating systems and software programming languages such as Microsoft's Windows operating system is a very common operating system. This means that this OS can have many different systems and applications which can do different jobs, making the PC an important piece of technology that can be used for almost every job within the company.

The Windows operating system is also compatible with many computer systems and it usually is the default OS for a computer, meaning that the PCs in a company can be much cheaper with a Windows OS when compared to a Mac.

Also, since the operating system has the ability to have a programming program, this means that there is a possible wide range of systems and software that can be made via programming that can be tailored to fit with many business needs, such as logistics.

<u>The estimated purchase price and running costs</u>

The price of a GPS that would be suitable for a logistics company can vary depending on the vehicle that the products are being transported in. This means that the overall estimated price cannot be said.

However, the estimated price for a GPS can be said depending per vehicle. For instance, a decent logistics GPS that can be installed in a delivery truck can cost around £300-400. While a decent GPS for a ship can cost around £700-1000.

The price of tracking applications that would be suitable for a logistics company can also vary depending the device it is being used in or whether it is a paid app or not. However, a paid app usually is better than a free application, this can be due to the fact that a free application would have advertisements or that it is very inefficient due to it being made with hardly any effort, causing it to be completely inexpensive.

For instance, a decent tracking application that can be installed within a PC could cost a monthly subscription which can range from £30 to £100. While a tracking application that is on a mobile device can cost a affordable price of £5 or even for free, but these are nowhere as efficient as the expensive PC applications.

The price of a portable device that would be suitable for a logistics company can also vary depending the device. What I mean by this is that each company that makes a mobile device makes them in different ways with different hardware, meaning that each piece of hardware can have a wide range of costs, but some are much better than others.

For example, a price of a tablet such as an Apple iPad would cost around £400-700. Furthermore, there are cheaper tablets available such as the Samsung Galaxy Tab S2 with a price of £200-300. However, this tablet has much less performance than the iPad, which can be more problem inducing.

The price of operating systems and programming languages, like the tracking applications, can depend on who made the operating system and programming language software, how expensive it is and how compatible the software is.

For example, Microsoft's Windows 10 Operating System's price can range from £90-100. This is an expensive OS due to it being relatively new, but it is widely used and extremely compatible to a lot of software when compared to other Operating Systems like Mac OS and Linux.

For a programming language software, Microsoft's Visual Studio is a very good program and it is a free to download program, meaning that this software would be very inexpensive to obtain. This means that the business won't have to dispose much money into a programming software, unless the business decides to go for a high end software which requires monthly or permanent payment.

Suggest the types of organisation that may also use the technology

DHL and other logistic companies alike aren't the only companies that can benefit from this type of technology. This is due to the wide potential that this technology can bring. For instance, organisations such as retail and manufacturing companies can benefit from this technology.

For retail companies, they can benefit from technology such as the ATS, portable devices and Operating systems. I feel that this technology would be useful for retail due to the business focusing on selling products, instead of producing them or supplying.

They benefit from this technology because it helps make sales much more efficiently, due to the orderly tracking of the staff and products, giving it a smooth process where an error can be conveniently corrected.

Basically, with the information and data of the staff sale rate and the amount of products, the company can find out which product sold the most and which sold the least. Which they can then use for sales to attract more customers to buy the product.

For manufacturing companies, like retail companies, they can benefit from technology such as the ATS, portable devices, Operating systems and with a GPS tracking system. I feel that this technology would be useful for retail due to the business focusing on producing products, instead of selling them or supplying.

They benefit from this technology because it helps the production of the business in a smooth flow due to the tracking of the work force and the parts used, meaning that everything goes where it needs to be, reducing the errors and misplacement of parts, products and staff.

Basically, with this data and information of the storage in the inventory and tracking how many there are and where they are, the company can organise the parts and products to go to the right area to the right time, making it less likely for the parts and products to get lost.

List at least two benefits the technology will bring to an organisation

One benefit that this technology will bring to DHL is that it would significantly increase the efficiency of the businesses flow. What I mean by this is that the tracking of staff and deliveries means that the company knows where the workforce is going and whether they are going to the right place.

This is a benefit because this gives the company the ability to monitor their staff and deliveries, gathering data that can be used to the company's advantage. For example, the company can gather the data of multiple supply vehicles that go to the same point to another point, tracking which route is the fastest. Which they will then use that data to increase the speed of deliveries, which can help improve the image of the company and the fuel efficiency of the vehicles.

Another benefit to DHL is that it helps give feedback to the companies and organisations that the logistics company is working for. What I mean by this is that if a company ordered a lot of products to be delivered, DHL would be able to tell the company where the delivery is and even when it will arrive.

This is a benefit because this information can be used to give to other companies so that they have an estimated time to get the supplies delivered. The reason that this is a benefit is because it makes the logistics company seem much more reliable when compared to the competition, so it is more likely that the company would be a recommended for delivery for another company or organisation in need for a logistic company.

List at least two drawbacks the technology may bring

One drawback that this technology would bring is that the technology that is used is not completely flawless. What I mean by this is that there might be technical difficulties that can occur, which can sometimes be fatal towards the logistics company and the the company or organisation that they are serving.

This is a drawback because the technology crashes or having a malfunction,it would impact the data flow, causing it to instantly stop, but that isn't the only issue. If it was a GPS that stopped working, the driver of the vehicle would not know where to go, causing the supply to be delayed.

If it was the tracker that stopped working, the business would not be able to monitor the workforce or the supplies, making the business blind on its progress. With other technology, such as the OS, tablets or other software, it will still have an impact.

However, it wouldn’t have as much as an impact because the database storage would most

likely be stored online or a cloud server, so no work would be lost, but it also doesn't mean that the cloud/online storage is completely safe too.

The online storage can sometime malfunction and have outages, which means that the company won't be able to access the data while it's in the downtime. The worst case scenario would be if all the data would be lost, which is why multiple backups are recommended to have within the information system.

Another drawback is that the technology can sometimes not provide sufficient information towards the company. What I mean by this is that there might be a problem in transferring data or that data that isn't useful is gathered, which means that the company can't use useful data for their company, which can sometimes leave the company lost.
This is a drawback because the lack of information could mean that the company would be gathering data that wouldn't be beneficial at all, making the data gathering technology a waste of time.

For example, the company might use GPS and trackers on the supply truck, but each truck takes a different journey, meaning that they cannot calculate the fastest possible time that it would take to reach a specific point, or even if the point was just a temporary point.

(U3:P4) - Describe the features and functions of information systems

Describe how that technology can be used in Management Information Systems (MIS)

This technology can be used in as management for information systems for MIS because of the ability to collect data that the technology uses, and using it on the business's database for financial information. The way the data is collected depends on the type of technology that is being used, but they all can collect data which can be used efficiently.

For the hardware, a GPS installed on every delivery vehicle can be used to calculate how many vehicles there are, and how many deliveries it made each day. This data can be used in an MIS database to calculate how many deliveries the company makes a day. However, each delivery varies in price and the GPS cannot store how much each delivery costs.

For the software, an applicant tracking system can be used to track the employees on the logistics company. This can give the company data on how many employees it has, how much they are paid, how many employees are joining, etc.

This data can be used for the business to control and maintain the employees and the finance to do with the company's employees, increasing the chance of promotion within employees and employees getting fair pay.

For the wireless devices, such as tablets, smartphones and smartwatches can be used to gather data depending on the role of said wireless device. For example, a tablet can be used for the employee to write in a report of the delivery. Saying how much it was and giving other information that a GPS or tracker can give, such as the busiest hours for traffic and whether there is a construction on the read.

For the operating systems and software programming languages, this technology can be used for the storage, access or even the creation of the MIS database. This is a very important piece of technology because it is the creation of the MIS, and without it the company would struggle to handle all the data at once.

Overall, technology is extremely important for MIS because technology handles data better than people do. SO with technology, you can create, gather, store and use the data for the benefit of the business much more efficiently than if the business had no technology at all.

Suggest if and how the technology may collect Data and generate useful Information.

For the logistics company,DHL, it would be a good idea for the company to collect the data that is gathered from the technology used within the business, using that data to generate useful information which can be stored within a database.

The highest priority for the business, is that database has contain information which can benefit the company in many different aspects. Such as finance, marketing, customer service, staff working hours and other useful information that help with efficiency of the deliveries.

The reason this is such a high priority for DHL, is because it is a critical point of a business that is too much of a risk to be left alone, due to the company's purpose. So, if that was left alone, there would be an extremely high chance that a problem may occur in that area and cause the downfall of the company.

However, this isn't the only area the business could use database for, even though it's the highest of priority, it doesn't mean that the other aspects should be left alone. Having data with other aspects such as the stocks of the company and similar finance data is also very important to have.

This is so that company can maintain the flow of money to the business, so it stays alive and has enough disposable income to spend on many benefits to the business,such as new and efficient technology, and the workers for the business.

Overall, the data that is gathered into a database can be extremely beneficial when used correctly. The data can help improve the business in places where they previously didn't notice due to not investigating in that general area and it also helps management of the company, financially and with the company's competitors.

Describe whether the technology's function is to input, store, process or output information

The technology's main purpose yet again depends on what type of technology it is, whether it is hardware, software, wireless or a operating system, they can all input, store and process or output information. However, each bit of technology can do few of these abilities, one piece of technology's handling of information is different to the other.

For the hardware, since a GPS is used to calculate how many vehicles there are, and how many deliveries it made each day, this means that the data from this technology is inputted and processed. Not only that, but the GPS can also process the information due to its automated

function.

The reason I think this is due to the fact that data is sent MIS database to calculate important information automatically. Basically, the data that a GPS can gather is only sent to the MIS to then be stored.

For the software, since an applicant tracking system is used to track the employees on the logistics company, it means that the data from this technology is also outputted. The reason I think this is due to the fact that data the software gathers is sent MIS database to calculate important information. Basically, the data that the applicant tracking system can gather is only sent to the MIS to then be stored.

For the wireless devices, since a tablet can be used for the employee to write in a report of the delivery, meaning that the information is both outputted and stored. The reason I think this is due to the fact that data that the tablet gathers is sent MIS database to calculate important information.

Not only that, but the tablet can also store the information in the cloud or storage space, which can later be used. Basically, the data that the tablet can gather can be stored within its storage system / cloud storage when can then be transferred to the MIS database.

For the operating systems and software programming languages, since this technology can be used for the storage, access or even the creation of the MIS database, it means that this type of technology can input, output, store and process data.

The reason I think this is due to the fact that a PC is a very versatile machine that can carry out many tasks, so it wouldn't be difficult for a PC to do all these aspects. Basically, the PC is mostly used to store data, but it can also input, output and process data through the ability for the PC to calculate data and to upload and download data.

Overall, each different piece of technology has different role on handling data, with PC being the most versatile. However, if it were only a PC it wouldn't be able to gather near as much data as the rest of the technology combined. Which is why all this technology is important to have in a company and is not at all obsolete.

(U3:P5) - Identify the information systems used in a specified organisation

Clearly identify the information systems that DHL may use in the specified case study

The information systems that DHL might use can range from is the GPS and tracker application software, but it also would be useful to use the other software, obviously. Say, if the company only had GPS and tracker software as their only information system, the company wouldn't be able to control its data well.

Without a doubt, the GPS software would be beneficial to DHL due to its ability to improve the inbound logistics performance, same-day parts deliveries, giving the ability to redesign the aftermarket logistics to drive down costs and carbon and improving the load logistics provider solution which creates a competitive advantage for other companies.

Also, The tracker application software is useful for DHL to have due to it helping in the improvement of dynamic inventory control, gives the company a collaborative platform that delivers a sustainable competitive advantage and it can make a collaborative supply chain to other companies.

Overall, the GPS and tracker application technology is the most important technology for a logistic company such as DHL. However, other technology is also really useful. What I mean by this is that with the other technology, the company is able to get a lot of useful data that can be used to fulfil these company needs and goals.

Task 2

(U4:M2) - Suggest how recent developments may improve a business system.

Briefly introduce the technology

The latest technology is undoubtedly improving rapidly, with some technology being twice or three times as fast from technology from 10 to 5 years ago. With this mass evolution of technology, it could mean that businesses using them can also be significantly faster than if the business didn't have the technology.

For instance, there are a lot of technology that have massively improved that would prove to be beneficial to a business system. Technology such as laptops, better performing PCs, much more accurate GPS and the ability to navigate through maps using a device, sometimes even at real time so other information such as road blocks can be seen.

Write a paragraph that explains the types of businesses that could also be improved. Give an example of an additional Company by name if necessary

Since there is a vast amount of technology, this gives the ability to extend to a variety of businesses that could need a benefit that technology could bring, such as production and retail businesses. For example, if a retail company like Apple have this type of this type of technology within their company, they have a very enhanced method of collecting data from their sales, workforce and products made which can be used to improve on places where it is now noticeably inefficient.

Write a detailed account that suggests at least five improvements the technology may bring to the business you have introduced

Having this type of technology within the business could bring many benefits that could increase the sale efficiency of the entire business. For example, management of important parts of a business such as the supply chain and the stock control is much more efficient with technology.

Not only could it improve on management, but also the development and delivery of products. Aspects such as service delivery and product development is significantly more efficient with newer technology, or just technology in general.

With the customer relationship management, the technology can improve this area for many reasons, but most is due to the data that is gathered through the database. What I mean by this is that the information that is stored within a database can have information of the customer.

Customer information is usually obtained via feedback that the business received from a survey or an online review of a product, and using that data to improve the said product or service that the business provides.

For the supply chain management, the technology, such as trackers which can be used to store information in a database, can improve this area with the data that is gathered through the database.

What I mean by this is that the information that is stored within a database can have information of each warehouse inventory and how many crates of products are in the warehouse, what are inside the crates and where the crates are headed to.

This gives the business the ability to maintain the supply chain in a neat and orderly management, which can greatly reduce the chance of the wrong supplement of products or the supplies from being lost without any knowledge of it being lost.

The product development efficiency is increased by technology due to the much more common and rapidly increasing development with robotic technology, which is already used by other large companies that are in the production or packaging business.

What I mean by this is that robots may seem like a disadvantage to an employee due to replacement, but to a business it is very advantageous. Robotic technology is much more precise and faster than the average employee, and the technology doesn't require a salary like an employee would.

Basically, the reason that this technology improves the product efficiency is due to the fast and precise production rate of robot technology when compared to an average employee. Which means one robot makes more more money for a business than a group of employees would.

The service delivery is much more accurate and efficient due to technology such as the GPS for tracking and mobile devices for communication and other uses that would benefit the delivery process.

What I mean by this is that the ability for the company to track and communicate with the delivery, they are able to manipulate the route of the delivery for faster travel or they could tell the driver that the destination has changes, reducing confusion.

Basically, technology which gives the ability to track and communicate with the delivery service gives the ability to have a much more efficient route and prevents from errors and confusions from happening. Which reduces the chance of the company being seen in a bad way from the customer.

With the management of people, whether it is in the business itself or customers, technology such as an MIS database or online surveys/feedback from customers and employees can help

with the management of people and their needs.

What I mean by this is that the information that is collected from customer/employee feedback can make the business change some aspects that make some customers/employees unhappy or an unnoticed problem within the business that they have noticed. However, If there is conflicting opinions on the feedback from people, the business would have to resort on the largest average to appeal to a larger audience.

Basically, the ability for the company to store information about problems within the business, the business is able to tailor themselves to fit the needs for the majority of customers and employees.

For the stock control and overall management is improved with technology because of the data that is gathered through the database. What I mean by this is that the information that is stored within a database can have information of the stocks and the overall information about the business itself.

What I mean by this is that, having data about the overall business and the stocks the business is making, it makes the business monitor how well it is doing in the market. With this ability the company can spot any drop in the stocks and other problems within the overall company, and try and pinpoint where and why the problem is happening, and how to address that said problem.

Basically, if the business has information about how they are doing with their profits and how they are doing in general. This makes them look at a problem that can occur a lot easier without having to worry about not knowing what is going wrong, instead the business could look at the previous data and information and find out the root of a potential problem.

In conclusion, technology can prove to be very useful for a business that focuses in logistics, such as DHL, making them much more able to maintain and fix potential problems within their overall business and stocks, which is undoubtibaly a critical importance for a business.